relAI

Konrad Zuse
School of Excellence
in Reliable AI

Bayesian Sample Inference for Graph Generation

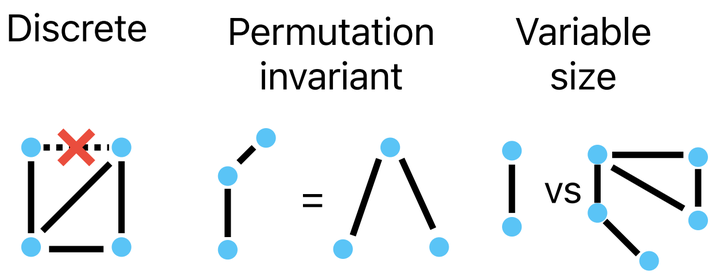
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Graph generation is challenging

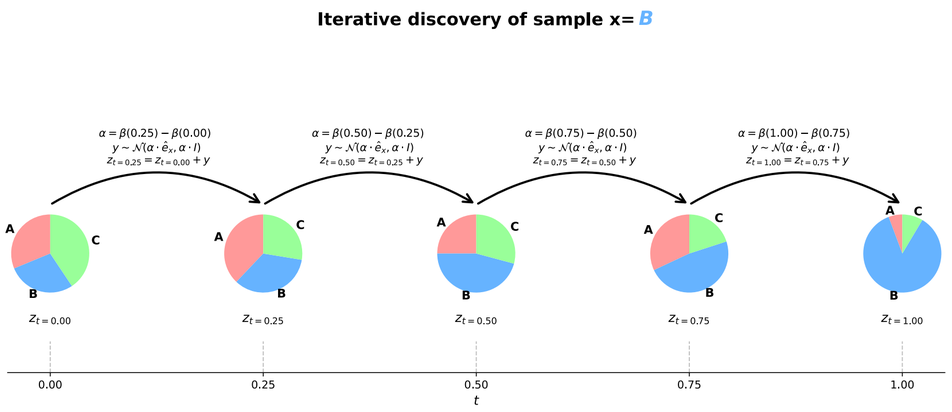


Bayesian Sample Inference (BSI) is diffusion in parameter space

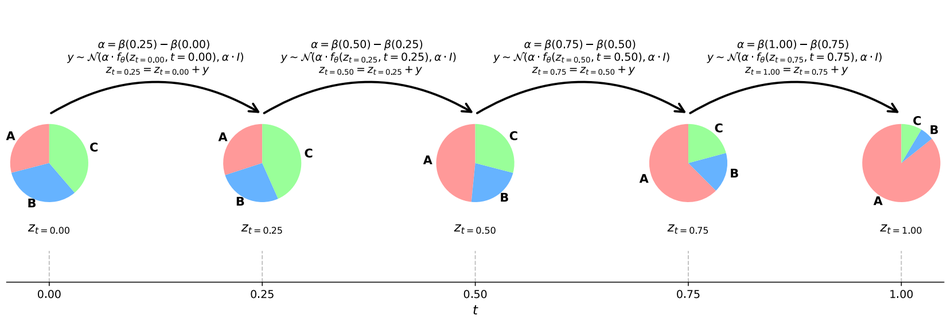
- Belief $p(\boldsymbol{x}|\boldsymbol{z})$ about sample $\boldsymbol{x}$ parameterized by latent variable $\boldsymbol{z}$:

$$p(\boldsymbol{x}|\boldsymbol{z}) = \mathcal{C}(\boldsymbol{x}|\mathrm{softmax}(\boldsymbol{z}))$$

- New information is gradually incorporated in the belief $\boldsymbol{z}$ through noisy observations $\boldsymbol{y}$ of $\boldsymbol{x}$ at precision α and Bayesian updates.



- Neural network f_θ is trained to predict train samples $\boldsymbol{x} \approx f_\theta(\boldsymbol{z_t}, t)$
- To generate new samples, the train sample $\boldsymbol{x}$ is replaced with a $f_\theta(\boldsymbol{z_t}, t)$ in every step:



Training and sampling with BSI is remarkably simple

Algorithm 1 Sampling with Categorical BSI

Require: reconstructor f_θ, discretization Δt, precision schedule $\beta : [0, 1] \to \mathbb{R}^+$
$z \sim \mathcal{N}(0, \beta_0 I)$
for $t = 0 \ldots 1$ in steps of Δt **do**
  $\hat{x} \leftarrow f_\theta(z, t)$
  $\alpha \leftarrow \beta(t + \Delta t) - \beta(t)$
  $y \sim \mathcal{N}(\mu = \alpha \cdot \hat{x}, \Sigma^2 = \alpha \cdot I)$
  $z \leftarrow z + y$
end for
$x \sim \mathcal{C}(\mathrm{softmax}(z))$
return x

Algorithm 2 Training Categorical BSI

while not converged **do**
  $x \sim p(x)$
  $z_0 \sim \mathcal{N}(0, \beta_0 I)$
  $t \sim \mathcal{U}(0, 1)$
  $\alpha = \beta(t) - \beta(0)$
  $y \sim \mathcal{N}(\mu = \alpha \cdot x, \Sigma^2 = \alpha \cdot I)$
  $z = z_0 + y$
  $\hat{x} = f_\theta(z, t)$
  $\mathcal{L} = \beta'(t)/2 \cdot ||\hat{x} - x||_2^2$
  $\theta = \theta - \eta \nabla_\theta \mathcal{L}$
end while

We generate node features and adjacency matrices

- n-node graph with c_V and c_E node / edge categories is one-hot encoded:

$X \in \{0, 1\}^{n \times c_V}$ and $E \in \{0, 1\}^{n \times n \times c_E}$

- Variable node count n by first sampling n from marginal, then generate n-node graph

BSI works great for molecule generation!

Model	Guacamol					MOSES						
	Val.↑	V.U.↑	V.U.N.↑	KL div↑	FCD↑	Val.↑	Unique.↑	Novelty↑	Filters↑	FCD↓	SNN↑	Scaf↑
Training set	100.0	100.0	0.0	99.9	92.8	100.0	100.0	0.0	100.0	0.01	0.64	99.1
DiGress	85.2	85.2	85.1	92.9	68.0	85.7	**100.0**	95.0	97.1	**1.19**	0.52	14.8
DisCo	86.6	86.6	86.5	92.6	59.7	88.3	**100.0**	97.7	95.6	1.44	0.50	15.1
Cometh	98.9	98.9	97.6	96.7	72.7	90.5	99.9	92.6	**99.1**	1.27	0.54	16.0
DeFoG (10% steps)	91.7	91.7	91.2	92.3	57.9	83.9	99.9	96.9	96.5	1.87	0.50	23.5
DeFoG	99.0	99.0	97.9	**97.7**	**73.8**	92.8	99.9	92.1	98.9	1.95	**0.55**	14.4
Ours (10% steps)	**99.3**	**99.3**	**99.2**	92.5	54.0	**99.8**	**100.0**	92.6	96.7	**0.92**	**56.3**	**82.2**

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relAI is supported by the DAAD programme Konrad Zuse Schools of Excellence in Artificial Intelligence, sponsored by the Federal Ministry of Education and Research.

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